BONE STORIES - EDUCATIONAL KIT

Activity 1
Find the astragalus

An investigation to identify the talus/astragalus in the foot, understand its place in the tarsus and transfer this landmark to other mammals.

AUDIENCE 8-12 years old	**DURATION** 30-60 mins
CONTENT Guide, sheet, supports, correction	**LENGTH** 16 pages

In this kit

1. Facilitator guide
2. Learner sheet
3. Observation support
4. Answer key and leads
5. Sources and credits

Pilot version - content to be tested and proofread before final release.

Recommended printing: guide and correction for adults; student sheet in one copy per student; supports to be projected or printed in color. The kit also works in grayscale. Double-sided possible, long edge reversal.

Version: 1.0 pilot - July 20, 2026.

BONE STORIES - ACTIVITY 1

Facilitator guide
Find the astragalus

Operational guide for conducting an observational survey in a classroom or museum.

TYPE Facilitator guide	**AUDIENCE** Supervising adults
DURATION 30-60 mins	**VERSION** Pilot 1.0

Pilot version - content to be tested and proofread before final release.

1. In brief

The students observe an isolated bone, describe its reliefs, look for its position in the foot then justify their location with at least two clues. The comparison with the sheep or the cow allows us to transfer the landmark without assuming that the members have the same shape.

2. Objectives

- Locate the talus/astragalus in the tarsus region.
- Describe a shape before naming it.
- Link position, joint surfaces and movement.
- Compare humans and other mammals with caution.

3. Expected success

Successful

The tarsus region is identified and two clues are explained.

In progress

The area is plausible, but only one clue is formulated.

To resume

The answer is based on name or chance, without observation.

4. Duration and material

Context	Duration	Material
Class	45-60 mins	Learner sheet, pencils, printed or projected materials.
Museum	30-45 mins	Display case, skeleton, photograph or authorized replica.

5. Quick preparation

À PRÉPARER
Print one sheet per student. Keep the fix out of view. Prepare the vocabulary: foot, tarsus, talus/astragalus, joint, clue.

À DIRE
“Describe first. Then search. Justify with two clues. »

6. Detailed procedure

Time	Adult action	Student action	Purpose
5 mins	Present the bone without giving its location.	Formulate initial ideas.	Bring out representations.
10 mins	Request a description without a name.	Note bumps, hollows, surfaces and symmetry.	Install observation.
10-15 mins	Have the tarsus region searched.	Circle an area and write two clues.	Locate and justify.
10 mins	Distribute animal views.	Compare position and organization of the limb.	Transfer the landmark.
5-10 mins	Discuss the findings as a group.	Write a cautious conclusion.	Distinguish observation and hypothesis.

7. Questions and reminders

To observe

- What do you see without naming the bone?
- Where are the smooth surfaces?
- Which parts seem to fit together?

To justify

- What detail supports your proposition?
- What is missing to be sure?
- Does your second clue confirm the first?

8. Responses or expected elements

- The talus/astragalus is located in the tarsus, between the leg and the rest of the foot.
- Its smooth articular surfaces indicate contacts with other bones.
- In other mammals, a comparable bone exists, but the limb and proportions differ.
- A good answer describes the clues before announcing the location.

9. Differentiation

Need	Adaptation
Help	Limit the search to the area between leg and foot; provide the word bank.
Lighten the writing	Allow an oral or dictated response to a peer.
Go deeper	Compare three mammals and discuss differences in posture.

10. Museum adaptation

Use the sheet as a short station in front of a window, a photograph, a skeleton or a replica. Ask to locate a region of the limb rather than an isolated bone. An image or replica does not entirely replace the original. No heritage object is handled without authorization and protocol from the institution.

11. Quick assessment

Produce an output sentence: "I locate the talus/astragalus in... because I observe... and...". Check the presence of a location and two clues.

12. Common mistakes

> **À NE PAS AFFIRMER**
> Don't say that the talus is the visible heel. Do not use a simple silhouette likeness as proof of identification.

- Confuse the talus/astragalus with the calcaneus.
- Show an area without justification.
- Assume that all mammals have a foot identical to ours.

13. Sources and limits

The views of the talus come from BodyParts3D / DBCLS. Animal supports are anatomical images or historical plates credited in the support. The LeConte 1922 plate is old and does not replace a validated modern anatomical diagram. This activity introduces a method of observation; it does not train in expert anatomical identification.

14. Current status and remaining validations

> **À VÉRIFIER**
> Sheet and materials ready for a supervised test. Final scientific and educational proofreading, testing with students and physical printing still necessary. A modern anatomical diagram validated by the project remains recommended.

Print plan

Each student: student sheet, all pages.

Adult: guide and correction, one copy.

Projection: observation support.

Color: useful for talus in red; possible grayscale with legends.

BONE STORIES - ACTIVITY 1

Learner sheet
Find the astragalus

TYPE Learner sheet	**AUDIENCE** 8-12 years old
DURATION 30-45 mins	**VERSION** Pilot 1.0

Pilot version - content to be tested and proofread before final release.

First name:

Date:

Class/group:

Your mission

Find where the talus/astragalus is hidden in the foot and explain your answer with visible clues.

Objective: learn to observe a bone, suggest a location and explain what makes you think that.

Material: this sheet, a pencil and the observation support.

1. Describe before guessing

1. Look at the bone from several angles.
2. Write at least three details: shape, bumps, hollows or smooth surfaces.

Word bank: rounded - hollow - bump - smooth surface - narrow - wide - joint.

2. Find your place

On the support, landmark the area between the leg and the rest of the foot. Circle where you would place the bone.

I observe

Write what is really visible.

I guess

Write where the bone might be.

3. Justify with two clues

clue 1:

clue 2:

4. Compare with another mammal

What is similar	What changes	What I still need to check

5. My final trace

Complete: “I locate the talus/astragalus in ____________. My two clues are ____________ and ____________. »

À OBSERVER

An observation describes what you see. A hypothesis offers an explanation or a location. The two are not the same thing.

BONE STORIES - ACTIVITY 1

Observation support
Find the astragalus

To project or print. Captions and credits must remain associated with the images.

TYPE Visual support	**AUDIENCE** Learners and groups
USAGE Projection or printing	**VERSION** Pilot 1.0

Pilot version - content to be tested and proofread before final release.

1. Observe the human talus

view side. The visible shape depends on the angle.

BodyParts3D / DBCLS, CC BY-SA 2.1 JP, via Wikimedia Commons.

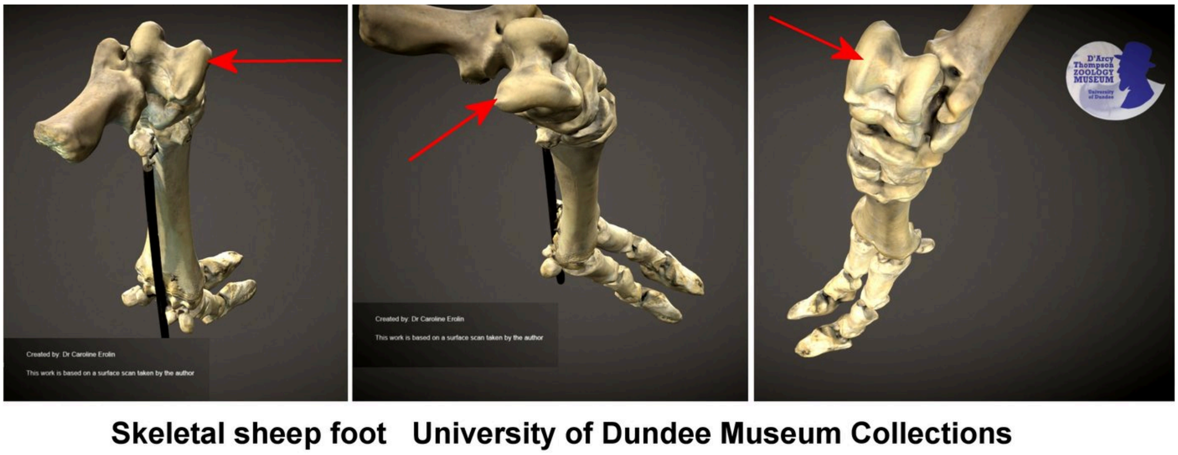
view anterolateral. Observe reliefs and joint surfaces.

BodyParts3D / DBCLS, CC BY-SA 2.1 JP, via Wikimedia Commons.

À OBSERVER

Red is only used to identify the bone in these visualizations. Also rely on the legends: the information remains readable in grayscale.

2. Switch to sheep



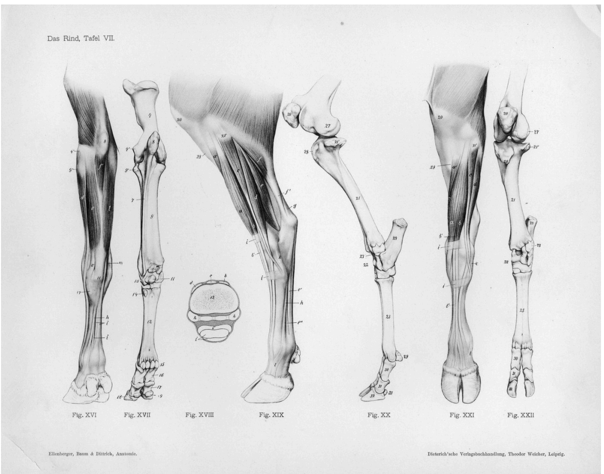
Annotated sheep's foot. Look for the tarsus region.

Caroline Eroline, CC BY-SA 4.0, via Wikimedia Commons.

Sheep skeleton. Replace the foot in the hind limb.

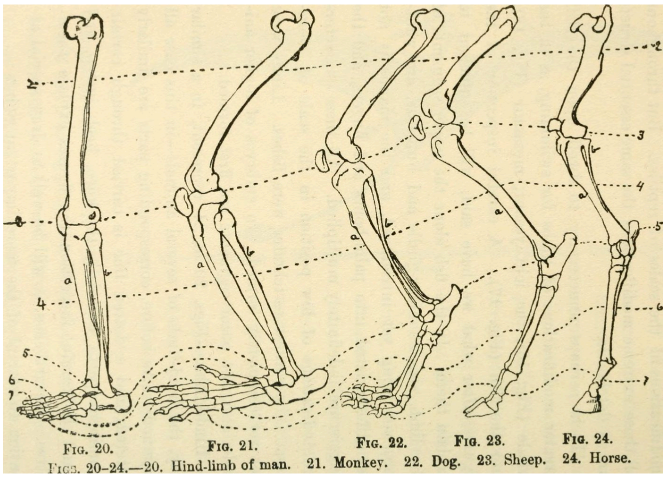
Museum of Veterinary Anatomy FMVZ USP / Wagner Souza e Silva, CC BY-SA 4.0.

3. Compare with caution



Old cow leg diagram. Comparison support, not modern project diagram.

Ellenberger, Baum and Dittrich, public domain, via Wikimedia Commons.



Comparative plate from 1922. Helps to observe; does not replace a validated modern anatomical reference.

Internet Archive Book Images, no known restrictions, via Wikimedia Commons.

BONE STORIES - ACTIVITY 1

Answer key and leads Find the astragalus

TYPE
Adult answer key

AUDIENCE
Teachers and museum educators

USAGE
Group review

VERSION
Pilot 1.0

Pilot version - content to be tested and proofread before final release.

landmarks fast

Form stage	Expected response	Possible answer	To resume
Describe	Three observable characteristics.	Rounded, smooth surfaces, hollows, reliefs.	“It’s a foot bone” without description.
Locate	Tarsus region, between leg and foot.	“Near the ankle, in the foot” if justified.	“In the visible heel” without nuance.
Justify	Two clues linked to the proposition.	Position and joint surfaces.	General resemblance only.
Compare	A similarity and a difference.	Comparable bone, different proportions.	“All feet are the same.”

Formulations adapted to students

Solid response

“I think the bone is in the tarsus because it is placed between the leg and foot and has several smooth surfaces that articulate with other bones. »

Cautious response

“This area is likely, but I would like to compare with a complete skeleton to verify. »

Common errors and correction

- **talus = heel visible:** remember that the talus is a deep tarsal bone.
- **One image is enough:** compare several views before concluding.
- **Similarity = proof:** request a second clue and a reference.

Scientific limits

The sheet allows an introductory anatomical identification. It does not replace either anatomical training or an osteological reference corpus. The talus/astragalus vocabulary is deliberately explained; the terminology depends on the anatomical, historical or archaeological context.

> **À VÉRIFIER**
> Have the final formulations reread, test comprehension by students and confirm the choice of the future anatomical diagram of the project.

BONE STORIES - ACTIVITY 1

Sources and credits

This page belongs to the complete kit. It allows you to use the documents without consulting the site.

Pictures

- talus human, views fixed: BodyParts3D / DBCLS, CC BY-SA 2.1 JP, via Wikimedia Commons.
- Sheep's foot: Caroline Eroline, CC BY-SA 4.0, via Wikimedia Commons.
- Sheep skeleton: Museum of Veterinary Anatomy FMVZ USP / Wagner Souza e Silva, CC BY-SA 4.0.
- Old cow leg diagram: Ellenberger, Baum and Dittrich, public domain.
- Comparative plate LeConte 1922: Internet Archive Book Images, no known restrictions.

landmarks scientists

- NCBI Bookshelf, *Anatomy, Bony Pelvis and Lower Limb: Foot talus*, landmarks general anatomical.
- BodyParts3D / DBCLS, anatomical visualizations used as localization support.

Status

Pilot version. credits integrated; scientific, educational proofreading and physical printing test still necessary. The 1922 plate remains clearly presented as an old document.

Working translation: specialist scientific, educational and legal wording requires human review.

Traceability : **Activity page, updates and online credits**.